

Article

Arrow of Time from Spontaneous Symmetry Breaking in Quantum Gravity

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Abstract: The question of the origin time's arrow is a major outstanding problem in physics. Here we present a possible mechanism for generation of a cosmological arrow via spontaneous symmetry breaking in a theory of quantum gravity. In [1], Chen and Vishwanath have shown that a local notion of time-reversal symmetry can be encoded into a gauge field living on a tensor network. Using the fact that tensor networks can be used to describe the state space of Loop Quantum Gravity [2,3] we show that a spontaneous symmetry breaking mechanism operating on such tensor networks can lead to the spontaneous generation of an arrow of time corresponding to the generation of non-zero order parameter for the Chen-Vishwanath gauge field.

Keywords: keyword 1; keyword 2; keyword 3 (list three to ten pertinent keywords specific to the article, yet reasonably common within the subject discipline.)

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The introduction should briefly place the study in a broad context and highlight why it is important. It should define the purpose of the work and its significance. The current state of the research field should be reviewed carefully and key publications cited. Please highlight controversial and diverging hypotheses when necessary. Finally, briefly mention the main aim of the work and highlight the principal conclusions. As far as possible, please keep the introduction comprehensible to scientists outside your particular field of research. Citing a journal paper [?]. And now citing a book reference [?].

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2.1. Subsection

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Bulleted lists look like this:

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- 33 • Second bullet
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- 36 1. First item
- 37 2. Second item
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39 The text continues here.

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41 All figures and tables should be cited in the main text as Figure 1, Table 1, etc.



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entry 1	data	data
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$$\S \tag{1}$$

44 Please punctuate equations as regular text. Theorem-type environments (including propositions,
45 lemmas, corollaries etc.) can be formatted as follows:

46 **Theorem 1.** *Example text of a theorem.*

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52 Authors should discuss the results and how they can be interpreted in perspective of previous
53 studies and of the working hypotheses. The findings and their implications should be discussed in
54 the broadest context possible. Future research directions may also be highlighted.

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Supplementary Materials: The following are available online at www.mdpi.com/link, Figure S1: title, Table S1: title, Video S1: title.

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Conflicts of Interest: The authors declare no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

MDPI	Multidisciplinary Digital Publishing Institute
DOAJ	Directory of open access journals
TLA	Three letter acronym
LD	linear dichroism

Appendix A

The appendix is an optional section that can contain details and data supplemental to the main text. For example, explanations of experimental details that would disrupt the flow of the main text, but nonetheless remain crucial to understanding and reproducing the research shown; figures of replicates for experiments of which representative data is shown in the main text can be added here if brief, or as Supplementary data. Mathematical proofs of results not central to the paper can be added as an appendix.

Appendix B

All appendix sections must be cited in the main text. In the appendixes, Figures, Tables, etc. should be labeled starting with 'A', e.g., Figure A1, Figure A2, etc.

References

- Chen, X.; Vishwanath, A. 'Gauging' time reversal symmetry in tensor network states. *Physical Review X* **2015**, *5*, [1401.3736].
- Qi, X.L. Exact holographic mapping and emergent space-time geometry, 2013, [1309.6282].
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